

Design and Performance Analysis of a Compact High - Efficiency Planar UWB Antenna for Wireless Applications.

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Abstract—Ultra-wideband communication systems require antennas that are compact, efficient, and capable of operating over a wide frequency range. In this study, a compact planar ultra-wideband (UWB) microstrip patch antenna is designed and analysed for wireless communication applications. The antenna is implemented on an FR-4 substrate with a dielectric constant of 4.4, a thickness of 1.6 mm, and overall dimensions of 26 mm × 26 mm. To improve impedance bandwidth and radiation performance, structural modifications are introduced in both the radiating patch and the ground plane. The antenna structure evolves from a conventional elliptical patch to a slot-modified configuration with a defected ground structure. The proposed design is modelled and simulated using CST Studio Suite to evaluate important antenna parameters such as return loss, voltage standing wave ratio (VSWR), gain, and radiation pattern. The simulation results show that the main resonance is at about 3.65 GHz and the return loss is -27.44 dB. At 6.38 GHz and 9.69 GHz, there are more resonances, which shows that the device can work in more than one band. The antenna keeps the VSWR below 2 at all operating frequencies and has a stable gain of about 3.8 dB. The structures of radiation also show that the performance is consistent across the frequency band that was studied. These results indicate that the proposed antenna is suitable for use in compact ultra-wideband wireless communication systems.

Keywords—Ultra-Wideband Band (UWB), Planar, Microstrip Antenna, Compact Design, Impedance Bandwidth, Antenna Gain, Radiation Efficiency

I. INTRODUCTION

The continuous technological growth of wireless communication has significantly increased the need in the creation of the antennas that will help to achieve a high speed of data exchange and reliable connectivity [4],[10]. The contemporary wireless devices require smaller antennas with good radiation characteristics and broad band of operation. With the increasing number of wireless applications, there is need to research on ways of designing small and functional antennas. The ultra-wideband (UWB) communication technology has received much coverage since it can transmit signals across a wide frequency range with minimal power [3],[6]. This renders UWB systems suitable in a wide variety of current applications that include high-speed, short-range communication, wireless sensor systems, indoor positioning,

and radar systems [11],[18]. Antennas that are used to support these applications should have a broad impedance band, strong radiation characteristics, and consistent operation in a series of frequency bands. Microstrip patch is currently ranked among the most popular category of antennas due to its simple manufacturability, inexpensiveness and compatibility with printed circuit technology [8],[12]. These are small, lightweight, and easily fitted antennas which fit in small electronic machines. Nonetheless, there are certain issues with the regular microstrip antennas, such as limited range, mediocre gain, and reduced radiation efficiency [13],[14]. To avoid these issues, researchers have proposed some design techniques to overcome them. Scholars have explored the issue of increasing the antenna spectrum and impedance matching, involving addition of slots, alteration of the ground plane, the use of parasitic elements, and experimentation with alternate feeding techniques [2],[7],[19]. It is these techniques that enable more resonant modes to be made and modify the flow direction of surface current which improves the wideband performance [16],[23]. In this paper, there are recommendations of a small, flat ultra-wideband patch antenna on microstrip. The antenna is constructed using 1.6 mm thick FR-4 substrate and is 26 mm x 26 mm in total. The structure is improved with time through the change of the shape of the patch and the defected ground framework. CST Studio Suite is the tool that we use to simulate the design and test of the key parameters of the antennas such as the return loss, VSWR, gain, and radiation patterns. The results suggest that the proposed antenna provides a variety of resonances in the working band and exhibits consistent radiation characteristics that suit the modern wireless communication networks.

II. THEORY AND PROPOSED WORK OF UWB MICROSTRIP PATCH ANTENNA

A. Antenna Theory

In the wireless communication system, microstrip patch antennas are widely used due to their small size and simplicity in manufacture [8],[12]. In a standard microstrip antenna three major components are used: a metal emitting patch, a dielectric substrate and a ground plane. This radiating patch usually comprises conductive materials such as copper and the substrate provides mechanical support to the antenna and interacts with the electromagnetic fields. On the other side of the substrate where the entire antenna structure was made are the ground plane. The emitting patches can be designed in various forms depending on the design requirements, which could either be in the

form of a rectangle, circle, triangle, and ellipse. Elliptic patch antenna is among the most widespread types of antennas as its shape is easy and its radiation characteristics can be predicted easily. In a microstrip antenna, electromagnetic energy is typically sent to the radiating patch by a microstrip transmission line feed. The reason this feeding method is popular is that it is easy to use and has been shown to be effective with planar antenna designs [5],[17]. Even though there are some good things about the microstrip patch antennas, there are also major limitations associated with the use of these antennas. Traditional design often shows a limited span of bandwidth of impedance and low radiation efficiency, which are limiting their attributes in wideband communication networks [13],[14]. These problems are even worse when the antenna is to be implemented in ultra-wideband mode where a significantly larger range of operational frequencies is to be supported. Structural changes, which may include adding slots to the patch, changing the ground plane, or changing the feeding structure can provide a significant improvement in bandwidth and impedance matching [2],[7]. These design changes allow the antenna to support a range of resonant frequencies by changing the current configuration in the antenna surface [16],[23].

B. Antenna Design

The proposed ultra-wideband antenna design is based on the 1.6 mm thick FR-4 dielectric substrate of relative permittivity 4.4. The general size of the substrate is 26 mm 26 mm which translates into a small planar fabrication which can be incorporated into modern day wireless devices. The radiating element of the antenna is an elliptical patch of microstrip that has been put on the upper surface of the substrate [6],[22]. It is a patch of approximately 23 mm by 23 mm. The antenna is driven through a microstrip transmission line feed due to its simplicity of implementation and construction [5],[17]. The feed line width of 3 mm was adopted to achieve an impedance of 50 Ω that will ensure the effective transmission of power between the source and the emitting patch. A defective ground structure (DGS) is introduced on the bottom of the substrate, and this increases the bandwidth properties of the antenna.

This modification improves the impedance matching of the antenna and changes the existing distribution in the ground plane. Also, the flawed ground enables the design to form multiple resonant frequencies, and this improves the wideband performance of the design [2],[14],[25]. Optimization and modelling of the antenna by CST Studio Suite results in a primary resonance of approximately 3.65 GHz with a return loss of 27.44 dB. The antenna has multi-band characteristics, and this is supported by the fact that there are other resonant frequencies at 6.38 GHz and 9.69 GHz. The optimized structure has also a maximum gain of approximately 3.8 dB over the entire working frequency range and similar radiation behaviour

TABLE 1. Design Antenna Parameters

Parameters Symbols	Ls	Ws	ts	Lp	Wp	Wf
Value(mm)	26	26	1.6	23.3	20	3
Parameters Symbols	Lf	Lg	Wg	tg	tp	
Value(mm)	12	12	26	0.035	0.035	

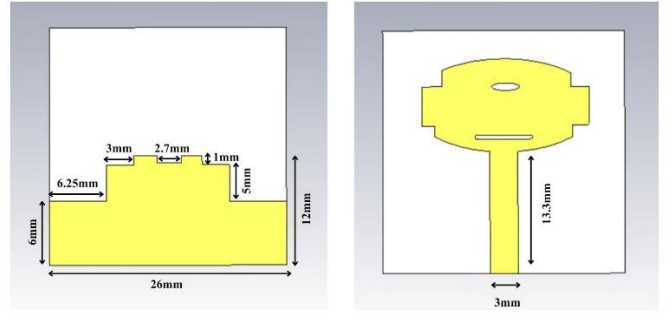


Figure 1: Final Optimized Antenna Structure

TABLE 2. Comparative Analysis of Design Stages

Step	Resonant Frequency (GHz)	Return Loss (dB)	Bandwidth Behaviour
S1	~ 3.8	-11 dB	Narrow
S2	~ 3.7	-16 dB	Improved
S3	~ 3.68	-21 dB	Wider
S4	3.65, 4.98, 9.69	-27.44 dB	Multi-resonant

III. SIMULATION RESULT AND ANALYSIS

CST Studio Suite was used to test the performance of the proposed ultra-wideband microstrip patch antenna. These data are analysed based on important parameters of the antennas, such as return loss, voltage standing wave ratio (VSWR), gain, radiation property and surface current distribution. Besides the simulated results, the behaviour is also observed in the light of the underlying electromagnetic principles.

A. Return Loss (S11 Parameter)

Return loss (S11) is another important parameter that is applied to measure the impedance matching between the antenna and the feeding transmission line. It shows the ratio of the power to the source because of an imbalance in impedance. To achieve effective working of the antenna, return loss is usually anticipated to not exceed 10 dB. The simulated S11 characteristics of the proposed antenna as demonstrated in figure 2 reveal that the antenna has a strong resonance at 3.65 GHz with a return loss of 27.44 dB, which is a strong indication of a good impedance matching. Further resonant frequencies are found at 6.38 GHz and 9.69 GHz, and these also testify to the multi-resonant nature of the design. Theoretically speaking, it can be explained that the multiple resonances are due to the alteration of the surface current pathways by the slot structure and the defected ground structure. Such structural changes enhance the effective electrical length of the antenna and more current modes are introduced and this makes it be operable in more than one frequency band [7],[16]. Such behaviour is necessary to attain ultra-wideband characteristics.

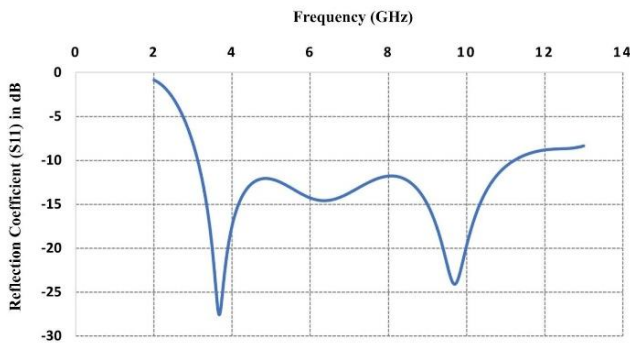


Figure 2: Graph Analysis of Final Optimized Antenna

B. Voltage Standing Wave Ratio (VSWR)

Another parameter that can be used to measure impedance matching is the voltage standing wave ratio (VSWR). It can be referred to as the ratio of the largest and smallest voltage on the transmission line. A value of VSWR that is near unity means that there are perfect matching and a value below 2 is usually good enough to use in practice. Figure 3 illustrates the simulated results of the VSWR show that the VSWR is less than 2 within the operating frequency range. This ascertains that the antenna has fine impedance that equals and effective transfer of power. It is possible to explain the observed VSWR performance by the optimized performances of the feed structure and geometry of the antenna [12],[21]. With the addition of the defected ground structure, the input impedance is altered and results in lowered reflection losses and thus enhances better matching conditions in a broader frequency range.

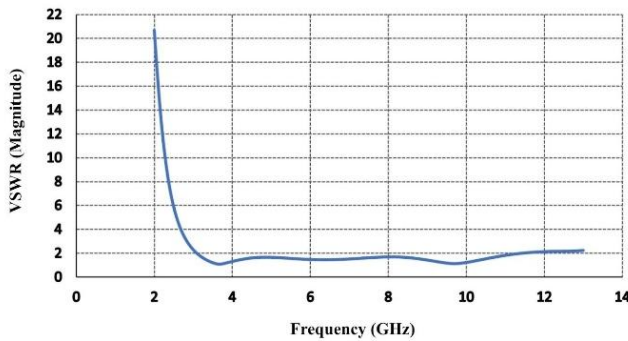


Figure 3: VSWR Graph Analysis

C. Gain and Radiation Pattern

The gain of the antenna is the capacity of the antenna to emit energy towards a desirable direction compared to an isotropic source. The proposed antenna is expected to attain an antenna gain of about 3.8 dB in the operating band as illustrated in figure 4. The gain over the frequency range is relatively constant pointing out to consistent radiation performance. This stability is significant to have consistent communication in ultra-wideband system. To give analysis of radiation pattern as depicted in figure 5, the antenna is almost omnidirectional in one plane where it has directional behaviour in the other plane. The pattern is appropriate when a uniform coverage of signals is needed. Theoretically, the nature of the radiation depends on the distribution of currents on the patch of the surface and the antenna geometry. The new introduction of slot and ground plane changes the existing distribution, which consequently, changes the radiation pattern and directivity

[16],[23]. Also, the dielectric losses introduced using an FR-4 substrate can cause a slight loss of the gain that can be attained [6],[22].

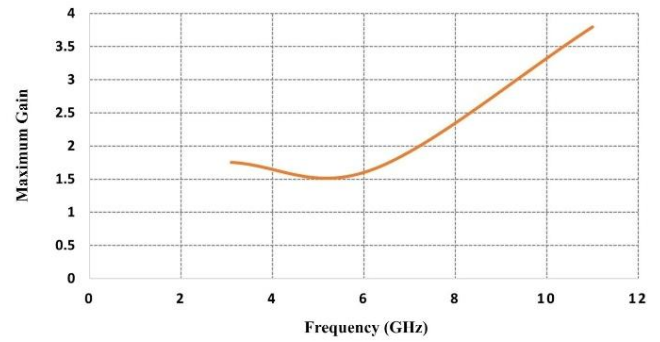


Figure 4: Maximum Gain Over Frequency

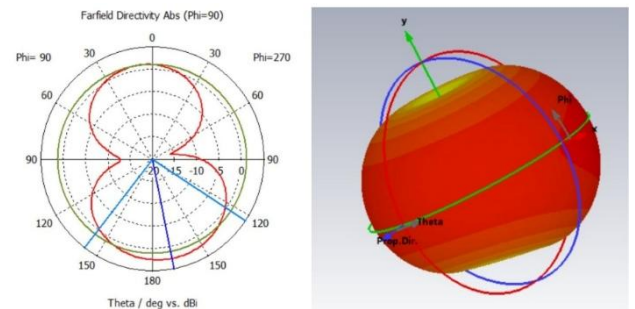


Figure 5: a) 3.1 GHz: 2D: Quasi-omnidirectional pattern with stable lobes. 3D: Broadside radiation with minimal distortion

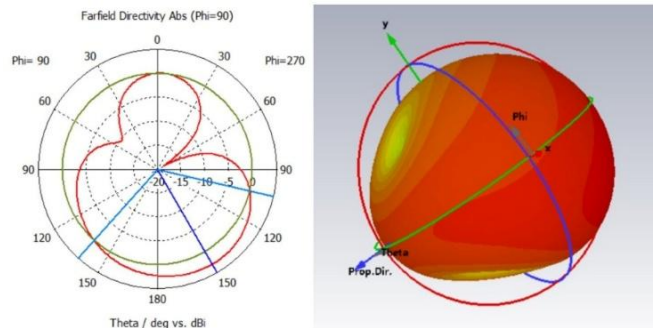


Figure 5: b) 6.2 GHz: 2D Plot: Stable pattern with minor ripples. and 3D Plot: Moderate dir. with uniform distribution

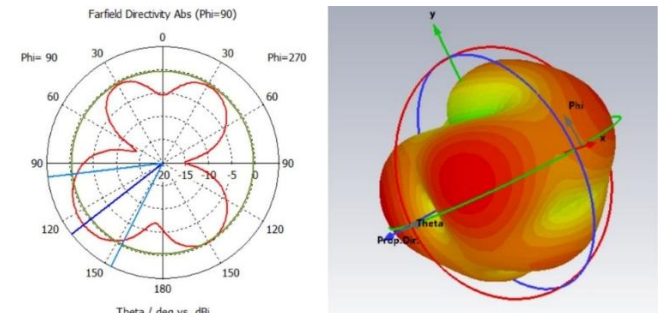


Figure 5: c) 11 GHz: 2D Plot: High directivity with slight sidelobes. and 3D Plot: Concentrated beam with control lobes.

TABLE 3. Comparison Of 2D Radiation Parameters

Parameters	3.1 GHz	6.2 GHz	11 GHz
Main lobe magnitude	2.56 dBi	2.87 dBi	1.55 dBi
Main lobe direction	169.0°	150.0°	128.0°
Angular width (3 dB)	94.0°	118.8°	56.5°
Side lobe level	-1.4 dB	-2.9 dB	-1.9 dB

TABLE 4. Comparison Of 3D Characteristics

Parameters	3.1 GHz	6.2 GHz	11 GHz
Type	Farfield	Farfield	Farfield
Approx.	$kR \gg 1$	$kR \gg 1$	$kR \gg 1$
Comp.	Abs	Abs	Abs
Output	Dir.	Dir.	Dir.
Rad. Eff.	-0.8072 dB	-1.221 dB	-1.613 dB
Total Eff.	-1.303 dB	-1.377 dB	-1.999 dB
Dir.	2.561 dBi	2.869 dBi	5.409 dBi

The radiation pattern is the distribution pattern of radiated energy of the antenna in space. The simulated radiation patterns at various angles of observation show that the antenna has almost an omnidirectional nature in one plane and a directional radiation pattern in the main plane. These radiation features render the antenna to be applicable in many wireless communication devices that need uniform coverage of their signals and other devices that need consistent performance.

D. Antenna Radiation Efficiency

The ratio of the power radiated by the antenna to the overall input power of the antenna is referred to as radiation efficiency. It is used to represent the quantum of the antenna in terms of its ability to convert electrical energy into radiated electromagnetic energy. The proposed antenna has a simulated radiation and total efficiency as illustrated in figure 6 and 7 respectively. The modelled performance shows that the radiation efficiency is not very sensitive to the operating frequency range. Even though FR-4 is reported to have a bigger dielectric loss than specialized substrates, the optimization of the antenna structure allows the antenna to achieve reasonable levels of efficiency. This is because the efficiency has been improved through better impedance

matching and optimization of current distributions. The defected ground structure is important in eliminating the unwanted reflections as well as improving the entire radiation performance of the antenna.

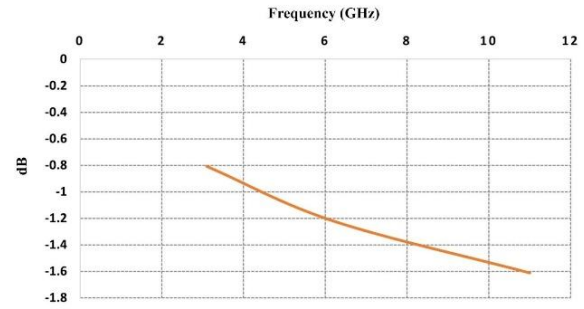


Figure 6: Radiation Efficiency

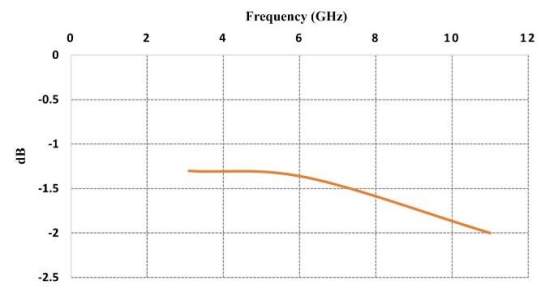


Figure 7: Total Efficiency

E. Surface Current Density Distribution on Antenna

Surface current distribution analysis gives good understanding of electromagnetic behaviour of the antenna at definite resonant frequencies. It assists in determining the areas of high current concentration that cause generation of radiations and resonance. Figure 8 shows the simulated distribution of surface current of the proposed antenna at 3.65 GHz which is resonance. The findings indicate a high level of current along the edges of the radiating patch, around the slot structure and along the microstrip feed line. Secondly, current distribution can be seen at the modified region of the ground plane prominently, and this shows that this area contributes to the improvement of the bandwidth. The existence of these areas of high current density contribute greatly to creation of resonance and enhancement of radiation properties. The comprehension verifies that slot-loaded patch and defected ground structure are effective in affecting the current distribution thus facilitating the bandwidth and the performance of the antenna [16],[23].

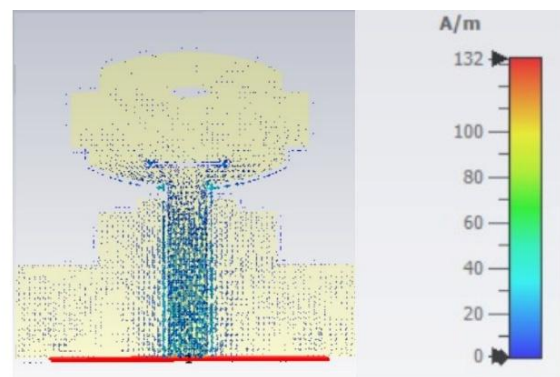


Figure 8: a) 3.1 GHz: Uniform current distribution

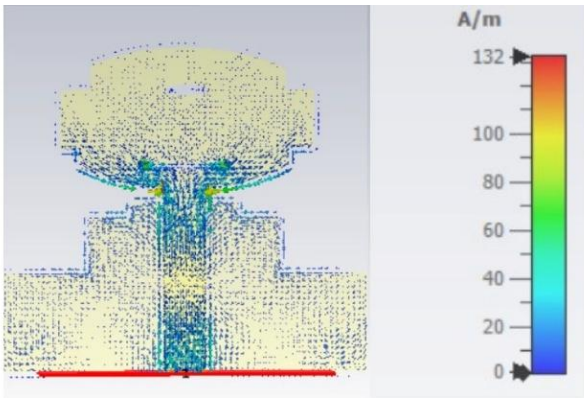


Figure 8: b) 6.2 GHz: Concentrated current near slots

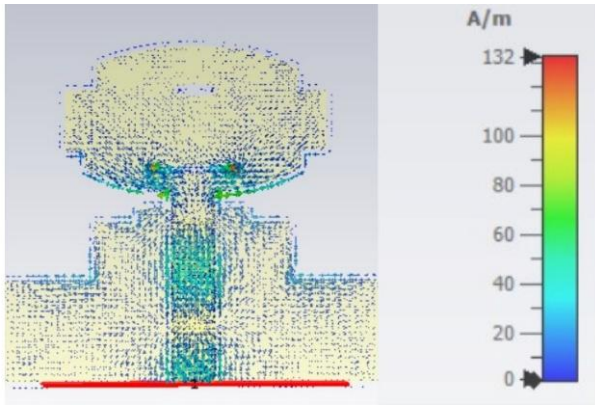


Figure 8: c) GHz: Strong current along edges

The general performance of the proposed antenna has revealed that structural adjustments like slot loading, and ground plane adjustment have a great impact on the electromagnetic behaviour of the antenna. It is possible to explain the improved performance through the following theoretical aspects:

- Enhancement of efficient electrical length with incorporation of slots.
- Multiple resonant modes were generated.
- Enhanced impedance matching by means of modification of ground plane.
- Surface current redistribution resulting in radiation characteristics improvement.

All these leads to the wideband performance and stable radiation behaviour of the antenna. The findings affirm that the design is good in small ultra-wideband wireless communications.

Advantage of Proposed Work

The ultra-wideband patch antenna in microstrip that has been developed has several viable benefits compared to the traditional microstrip design. The addition of a slot-loaded patch and defected ground structure better's impedance matching and allows the structure to be used at a variety of resonant frequencies in the desired band [2], [19]. The antenna can retain constant radiation features at reasonable gain functions over the working frequencies. The design is compact (26 mm x 26 mm) which makes it fit in space limited wireless devices. Besides, simple microstrip feed and planar structure are used which makes it very easy to make and compatible with standard printed

circuit technology. Thanks to these characteristics, the antenna may be regarded as applicable to the use in the ultra-wideband communication systems, short-range wireless connections, and IoT-driven devices.

IV. CONCLUSION

A miniature ultra-wideband (UWB) microstrip patch antenna has been developed and the antenna examined in this paper to be used in wireless communication. The design of the antenna was based on FR-4 substrate of small size 26 mm x 26 mm. This was done by implementing structural changes on the radiating patch and ground plane which resulted in better impedance matching and multi-band behaviour. The results of the simulation indicate that the main resonance is at 3.65 GHz, and the return loss is 27.44 dB. There were also other resonances at 6.38 GHz and 9.69 GHz which denoted the ability of the antenna to resonate in many frequencies. The antenna has a VSWR of less than 2 within the operating range providing effective transfer of power. It gave a stable gain of almost 3.8 dB and steady radiation properties. All in all, the findings suggest that the presented antenna offers an effective and compact design of the ultra-wideband wireless communication applications. The design is of good size, performance, and simplicity, which makes it applicable to implement the design in real world communication systems.

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